NeuroPulse AI: Electromyography (EMG) Profile

Yellow = cleaned signal | Green = muscle strength | Red = muscle active shadow

Patient: Patient Name | **ID:** NP-20260522-6079 | **Date:** 22-05-2026 19:45

Avg Strength	Active Time	Major Spikes	Signal Quality
0.1	**0.1%**	**1**	**Low**

1. Clinical Summary

Session Score	28/100
Symmetry Score	0.0%
Side Deficit	100.0% (Right lower)
Lowest Activity Channel	Channel 3
Fatigue Drop	Low (0.0% decline)
Signal Quality	Low

2. 30-Second Graph Snapshot

3. Muscle-wise Quantitative Summary

Muscle / Channel	Avg	Peak	Active %	Quality
Channel 1	0.4	70.8	0.6%	Low
Channel 2	0.0	8.0	0.0%	Low
Channel 3	0.0	0.0	0.0%	Low
Channel 4	0.0	0.0	0.0%	Low

4. Local Data-Based EMG Findings

Signal Reasoning (Quantitative Text Analysis):

* Session Score: 32/100. Signal Quality: Low.
* Side Deficit: 100.0%. Symmetry Score: 0.0%. Marked right-side deficit detected.
* Lowest Activity: Channel 3. Highest Peak: Channel 1.
* Fatigue Drop: Low (0.0% decline from early to recent activity).
* Channel Balance: left-side channel balance 50%; right-side channel balance 0%.
* Suggested focus: compare the two right-side muscles/channels and train the weaker activation pattern.
* Selected affected side: Select Affected Side. Non-diagnostic rehab monitoring support only.

Recorded Session Findings:

* Recording summary: 6570 EMG samples captured over approximately 60 seconds.
* Session Score: 28/100. Signal Quality: Low.
* Side Deficit: 100.0%. Symmetry Score: 0.0%. Impression: marked right-side activation deficit. Bilateral EMG symmetry is reduced and requires targeted rehab attention.
* Lowest Activity: Channel 3 showed the lowest average activation (0.0).
* Highest Peak: Channel 1 showed the highest peak activation (70.8).
* Channel Balance: left-side channel balance 9.7%; right-side channel balance 0.0%.
* Fatigue Drop: Low fatigue pattern with 0.0% decline from early to recent activity.
* Sustained activation: Channel 1 0.6%, Channel 2 0.0%, Channel 3 0.0%, Channel 4 0.0% above threshold.
* Clinical note: Affected side was not selected; interpretation is limited to channel activity and symmetry. This is a non-diagnostic rehab monitoring prototype that provides quantitative EMG feedback for physiotherapy support.

5. Clinical Observations & Notes

Summary Notes: N/A

Comparative Analysis:
- Left Side Avg: 0.2
- Right Side Avg: 0.0
- Symmetry Ratio (L/R): 0.0%
- Most Dynamic Channel: Channel 1